

Program 1

```
public class OddEvenNumber {  
    public static void main(String[] args) {  
  
        for(int i = 0; i <= 100; i++){  
            if(i % 2 == 0){  
                System.out.println(i + " is an even number");  
            }  
            if (i % 2 == 1) {  
                System.out.println(i + " is an odd number");  
            }  
        }  
    }  
}
```

Program 2

```
public class Main {  
    public static void main(String[] args) {  
        for(int i = 0; i <= 100; i++){  
            if(i > 0) {  
                if (i % 3 == 0 && i % 5 == 0) {  
                    System.out.println(i + " is divisible by 3 and 5");  
                }  
                if (i % 4 == 0 && i % 6 == 0) {  
                    System.out.println(i + " is divisible by 4 and 6");  
                }  
            }  
        }  
    }  
}
```

Program 3

```
import java.util.Scanner;

public class Palindrome {
    public static void main(String[] args) {
        System.out.print("Enter String: ");
        Scanner scan = new Scanner(System.in);
        String str = scan.nextLine();
        char [] ch = str.toCharArray();

        int num = 0;

        for(int i = 0; i < ch.length/2; i++){
            if (ch[i] != ch[ch.length - 1 - i]){
                num = 1; break;
            }
        }

        if(num == 0){
            System.out.println(str + " is Palindrome");
        }
        else{
            System.out.println(str + " is not Palindrome ");
        }
        System.out.println("character count: " + ch.length);
    }
}
```